Deep Learning

Deep Learning is a subset of Machine Learning that uses artificial neural networks with multiple layers.

Neural Network Components:
- Input Layer
- Hidden Layers
- Output Layer

Important Concepts:
- Activation Functions (ReLU, Sigmoid, Tanh)
- Backpropagation
- Loss Function
- Optimizer (SGD, Adam)

Types of Neural Networks:
1. CNN (Convolutional Neural Network)
   Used for image processing.
   Applications: Image classification, object detection.

2. RNN (Recurrent Neural Network)
   Used for sequential data.
   Applications: Text generation, speech recognition.

3. LSTM
   A type of RNN that solves the vanishing gradient problem.

Challenges in Deep Learning:
- Overfitting
- Vanishing Gradient
- High computational cost